Data Handling and Visualization

ECDF, Histograms, Density Plots & Q-Q Plots — Worked Solutions

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Question 1 — ECDF of Quiz Marks

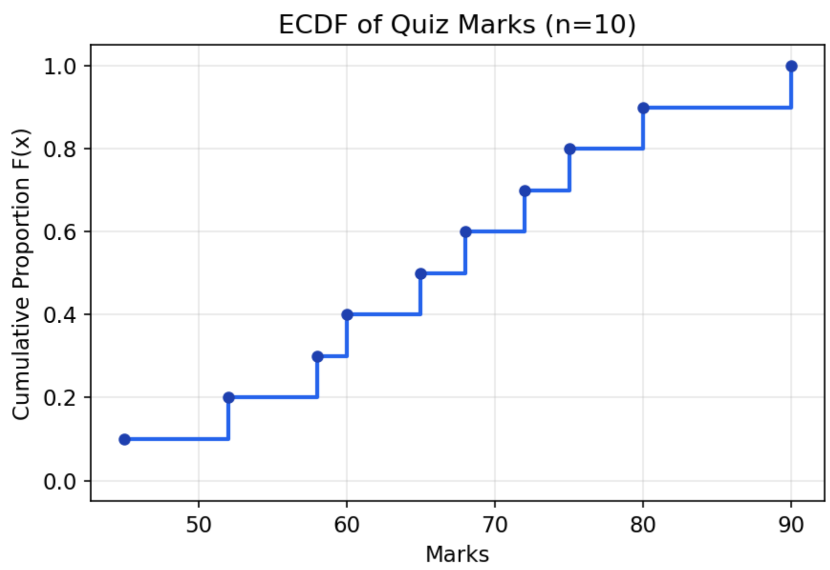
The Empirical Cumulative Distribution Function (ECDF) shows, for every value x, the proportion of observations that are less than or equal to x. For a sample of size n, each sorted observation increases the ECDF by 1/n.

Step 1: Sort the data and assign ranks

Formula: $F(x) = (\text{number of observations} \leq x) / n$, where $n = 10$

Rank (i)	Marks (sorted)	F(x) = i/10
1	45	0.10
2	52	0.20
3	58	0.30
4	60	0.40
5	65	0.50
6	68	0.60
7	72	0.70
8	75	0.80
9	80	0.90
10	90	1.00

ECDF Plot



Interpretation

- The plot is a non-decreasing step function that rises by 0.10 (1/10) at every data point.
- 50% of students scored 65 marks or below (median ≈ 65); 90% scored 80 or below.

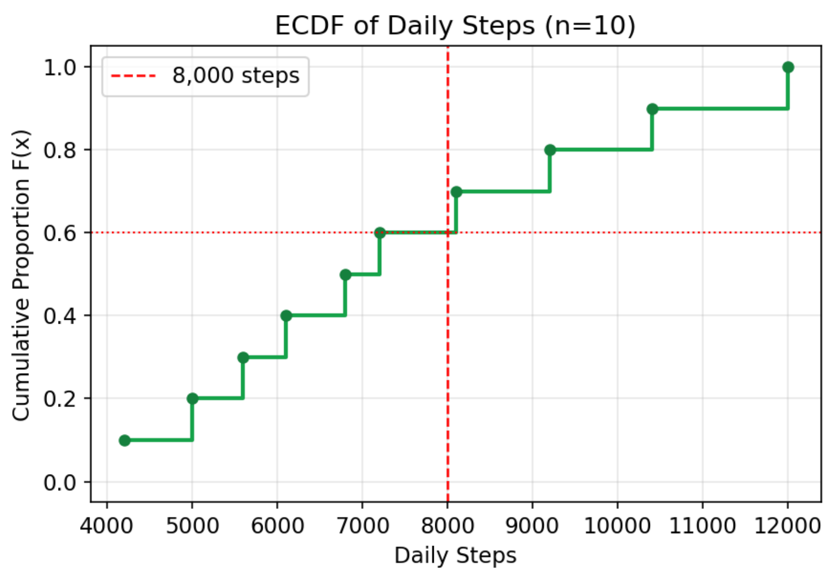
- The steps are fairly evenly spaced, indicating the marks are roughly uniformly spread across the 45–90 range with no large gaps or clusters.
- The ECDF lets us directly read off percentile-style questions, e.g. "what fraction scored below 70?" → 0.6 (6 out of 10).

Question 2 — ECDF of Daily Steps

Step 1: Sort the data and compute cumulative proportions

Rank (i)	Steps (sorted)	F(x) = i/10
1	4200	0.10
2	5000	0.20
3	5600	0.30
4	6100	0.40
5	6800	0.50
6	7200	0.60
7	8100	0.70
8	9200	0.80
9	10400	0.90
10	12000	1.00

ECDF Plot



Proportion who walked 8,000 steps or fewer

Participants with steps ≤ 8000: P1(4200), P2(5000), P3(5600), P4(6100), P5(6800), P6(7200) → 6 participants.

Proportion = 6/10 = 0.60 → 60% of participants walked 8,000 steps or fewer (equivalently, 40% exceeded the 8,000-step target).

Significance of the ECDF

- It gives an exact, assumption-free read of what fraction of the group meets any activity threshold (e.g., an 8,000 or 10,000 step goal), without needing to assume steps are normally distributed.

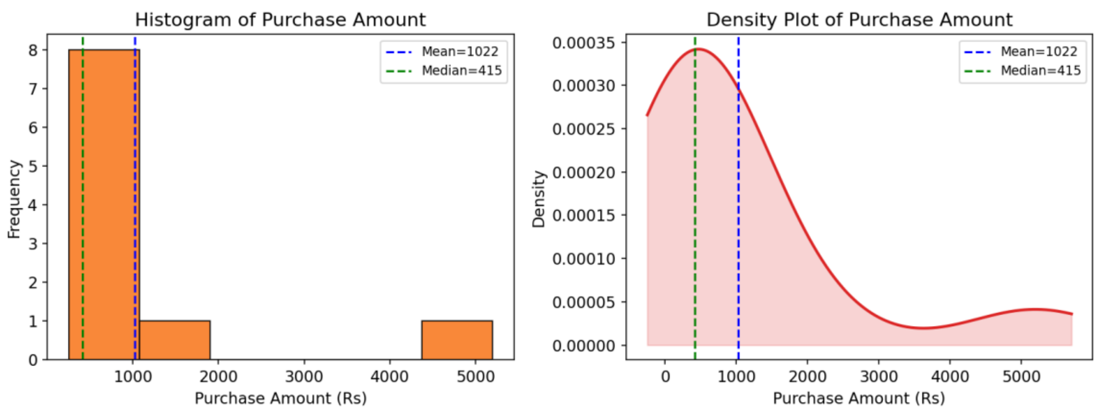
- Because it uses every raw data point (not binned like a histogram), no information is lost and no bin-width choice affects the shape.
- The visible larger jump between 8,100 and 12,000 (P7–P10) shows a handful of highly active participants pulling the tail to the right.

Question 3 — Distribution Shape of Purchase Amounts

Step 1: Summary statistics

Mean = ₹1,022.0   Median = ₹415.0   Sample Skewness ≈ +2.73

Histogram and Density Plot



Skewness Determination

- Mean (₹1,022) is much larger than the Median (₹415) → Mean > Median.
- The histogram shows a tall cluster of low purchase values (₹250–₹650) and a long thin tail stretching toward ₹5,200.
- The density curve is concentrated on the left with a long right-hand tail, and the skewness coefficient (+2.73) is strongly positive.

Conclusion: The distribution is Positively (Right) Skewed.

Justification: Most customers made small, inexpensive purchases (clustered near ₹250–₹650), while a few customers (C9 = ₹1,800 and especially C10 = ₹5,200) made very large purchases. These few high values stretch the right tail and pull the mean above the median — the classic signature of positive/right skew.

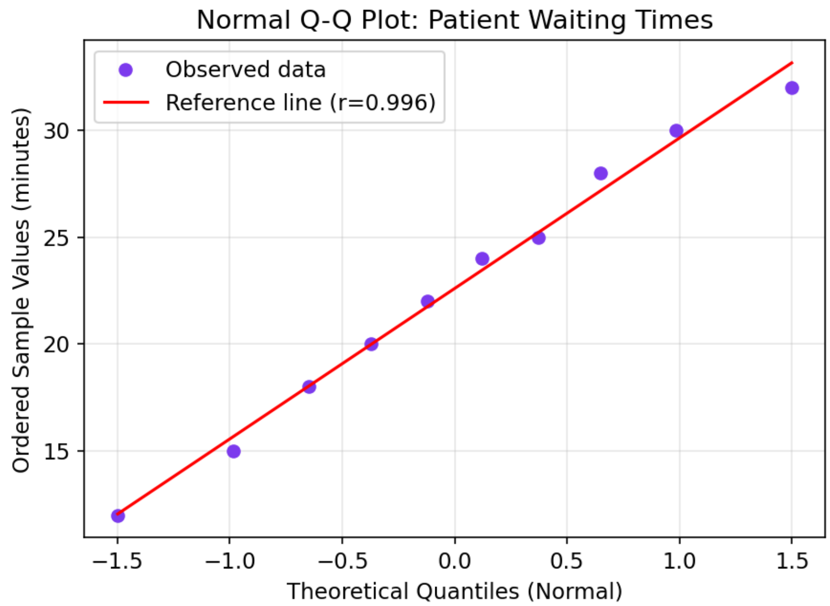
Question 4 — Normal Q-Q Plot of Waiting Times

Step 1: Summary statistics

Mean = 22.6 minutes   Std. Dev. = 6.48 minutes   Skewness ≈ −0.17 (nearly symmetric)

Normal Q-Q Plot

A Q-Q plot compares the sorted (observed) data quantiles against the theoretical quantiles of a Normal distribution. If the data is normal, points fall approximately along the straight 45° reference line.



Interpretation

- All 10 points lie very close to the straight reference line, with no systematic curvature or outliers at either end.
- The correlation between sample and theoretical quantiles is $r \approx 0.996$, extremely close to 1.
- The skewness (−0.17) is close to 0, consistent with an approximately symmetric distribution.

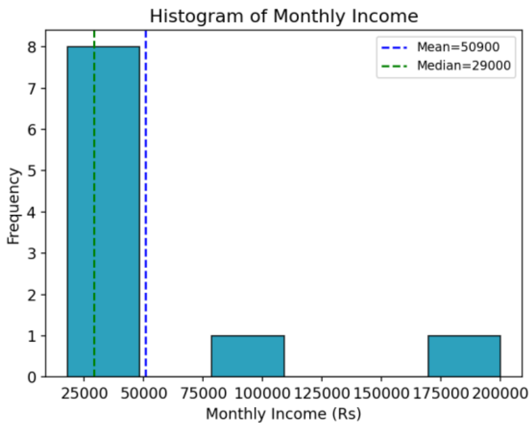
Conclusion: The waiting time data approximately follows a Normal Distribution — the assumption of normality is reasonable for this sample.

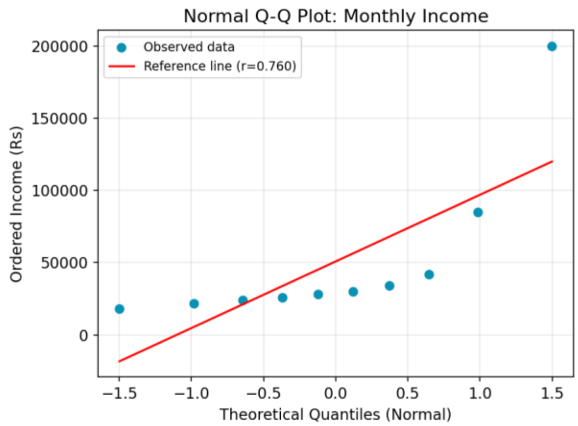
Question 5 — Monthly Income: Histogram & Q-Q Plot

Step 1: Summary statistics

Mean = ₹50,900   **Median = ₹29,000**   **Sample Skewness ≈ +2.59**

1. Histogram of Monthly Income   &   2. Normal Q-Q Plot





3. Normal or Highly Skewed?

- Mean (₹50,900) is far greater than the Median (₹29,000), a strong indicator of positive skew.
- The histogram shows most incomes clustered in the ₹18,000–₹42,000 range, with two extreme outliers (C9 = ₹85,000, C10 = ₹200,000) forming a long right tail.
- Skewness ≈ +2.59, far from 0, confirming strong positive (right) skew.

Conclusion: The income data is Highly (Positively) Skewed, NOT normally distributed.

4. How the Q-Q Plot Supports This Conclusion

- In the Q-Q plot, the lower/middle income points (up to about C8 = ₹42,000) lie reasonably close to the reference line.
- However, the top two points (C9 and especially C10) bend sharply upward, well above the straight line — this upward curvature at the right tail is the classic Q-Q signature of positive/right skew.
- The correlation coefficient (r ≈ 0.76) is noticeably lower than in Question 4, confirming a poorer fit to normality.

Together, the mean-median gap, the right-skewed histogram shape, and the upward-curving tail in the Q-Q plot all consistently point to a highly skewed (non-normal) income distribution — typical of income/wealth data where a small number of high earners pull the distribution's tail to the right.

Note on methodology: All ECDF values use F(x) = i/n (i = rank, n = 10). Skewness uses the sample (bias-corrected) formula. Q-Q plots use scipy.stats.probplot against a Normal reference distribution — the same statistical logic as R's qqnorm()/qqline() or Excel's rank-based normal-probability plots.